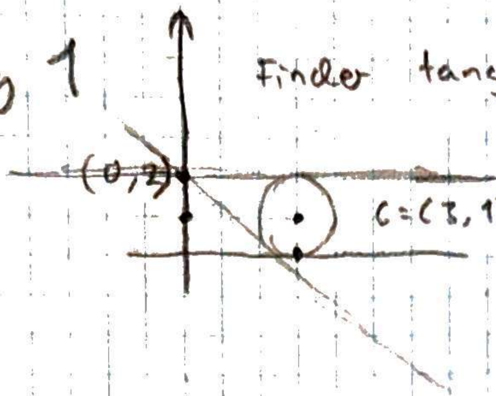
Opg 1



Finder tangenter

$$1 = \frac{|a\cdot 3 + 2 - 1|}{\sqrt{a^2+1}}$$

$$1 = \frac{(a\cdot 3 + 1)^2}{a^2+1}$$

$$a^2 + 1 = 9\cdot a^2 + 6a + 1$$

$$0 = 8a^2 + 6a$$

$$0 = a\cdot\left(a + \frac{6}{8}\right)$$

$$\underline{a = 0} \quad \vee \quad \underline{a = -\frac{6}{8}}$$

Opg 2

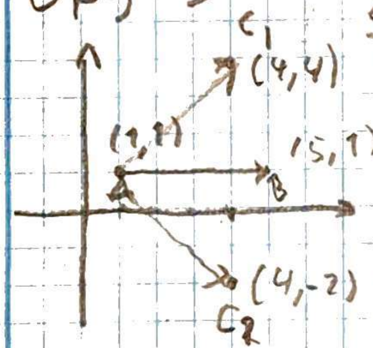
$$x^2 + y^2 - 4x - 2y = 4$$

$$(x - 4x + 4) + (y^2 - 2y + 1) = 4 + 4 + 1$$

$$(x-2)^2 + (y-1)^2 = 3^2$$

$$\underline{C = (2,1)} \quad \text{og} \quad \underline{R = 3}$$

Opg 3



Som det ses er

$$\overrightarrow{AC_1} = \begin{pmatrix} 4-1 \\ 4-1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$$

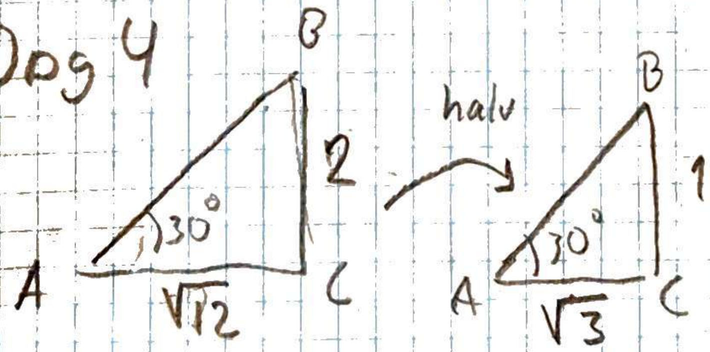
$$\overrightarrow{AC_2} = \begin{pmatrix} 4-1 \\ -2-1 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \end{pmatrix}$$

$$\overrightarrow{AB} = \begin{pmatrix} 5-1 \\ 1-1 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$$

$\overrightarrow{AC_1}$ og $\overrightarrow{AC_2}$ har vinklen $45°$ og $-45°$ ift x.

derfor $\underline{[45° ; -45°]}$

Opg 4



$$\frac{\sin(30)}{1} = \frac{\sin(B)}{\sqrt{3}}$$

$$\Updownarrow$$

$$\frac{\sqrt{3}}{2} = \sin(B)$$

$$\Updownarrow$$

$$\underline{B = 60°}$$

$$\underline{B = 180° - 60° = 120°} \text{ eller}$$

Opg 5 Reducer udtrykket

$$\frac{5\cdot(3^2\cdot 10)^2}{3^2\cdot 60^2} = \frac{5\cdot 3^{\cancel{4}\,2}\cdot \cancel{10^2}}{\cancel{3^2}\cdot 6^2\cdot \cancel{10^2}} = 5\cdot\left(\frac{3}{6}\right)^2 = \underline{\underline{\frac{5}{4}}}$$

Opg 6 Reducer udtrykket

$$\left(\frac{\sqrt{2}\cdot\sqrt{3}}{\sqrt{12}}\right)^6 = \left(\sqrt{\frac{6}{12}}\right)^6 = \frac{1}{2}^{\frac{6}{2}} = \underline{\underline{\frac{1}{8}}}$$

Opg 7

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2-1 \\ 3-1 \end{pmatrix}\cdot t + \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\underline{\underline{\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}\cdot t + \begin{pmatrix} 1 \\ 1 \end{pmatrix}}}$$

Opg 8

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}\cdot 0 + \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}, \quad t = 0$$

dvs. $\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}\cdot 1 + \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \quad t = 1$$

$$\begin{pmatrix} r_1 \\ r_2 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \end{pmatrix}$$

$$\underline{\underline{\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} + t\begin{pmatrix} -1 \\ -2 \end{pmatrix}}}$$

Opg 9 $\vec{n_L} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ punkt $(2,1)$

$L: \quad 1 \cdot x + 2 \cdot y + C = 0$

$1 \cdot 2 + 2 \cdot 1 + C = 0 \Leftrightarrow C = -4$

$M \quad \vec{n_m} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$ punkt $(-2,2)$

$-2 \cdot -2 + 3 \cdot 2 + C = 0 \Leftrightarrow C = -10$

$L: \quad x + 2y - 4 = 0$

$M: \quad -2 \cdot x + 3y - 10 = 0$

$M + 2 \cdot L: \quad +7y - 18 = 0$

$\underline{\underline{y = \frac{18}{7}}}$

indsat i L:

$x + \frac{18 \cdot 2}{7} = 4 \Leftrightarrow x = \frac{28 - 36}{7}$

$\underline{\underline{x = \frac{-8}{7}}}$

Opg 10

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos(t) \\ \sin(t) \end{pmatrix}, \quad t = [0; 2\pi[$$